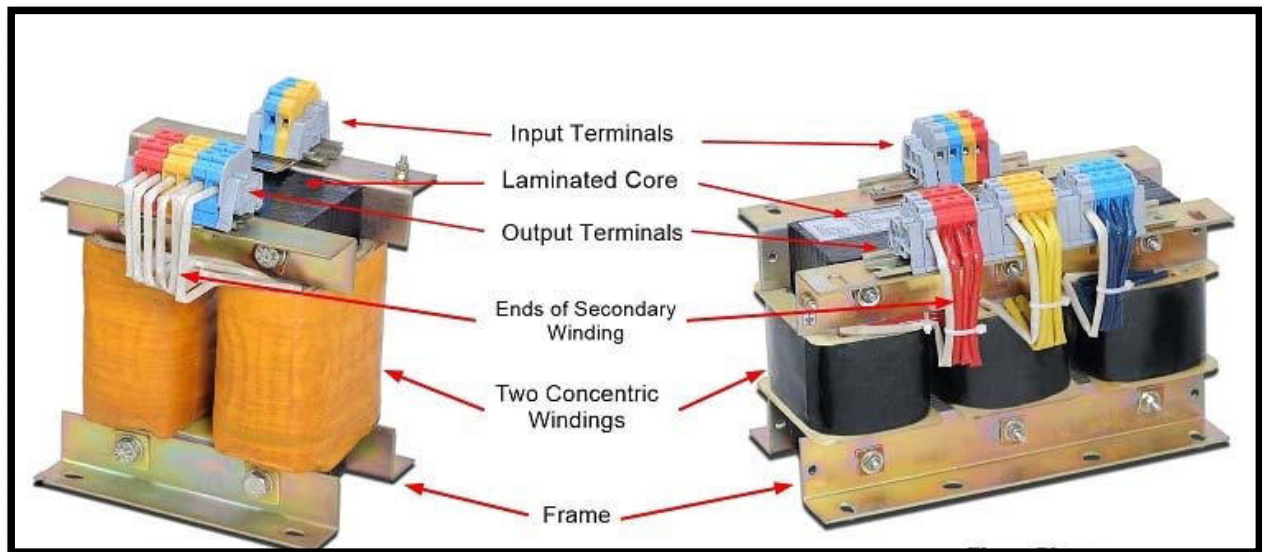


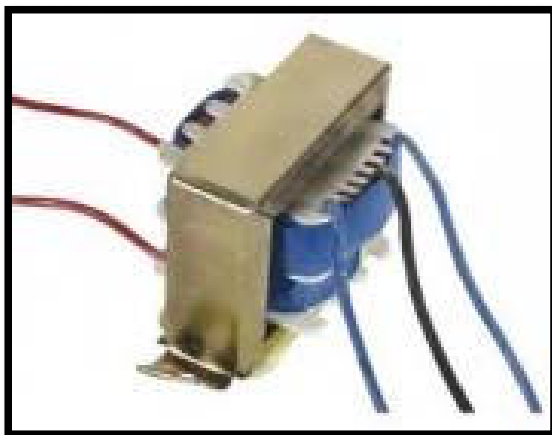
Core Type Transformer



Single Phase

Three Phase

Shell Type Transformer



Single Phase



Three Phase

EMF Equation of a Transformer

Step:-1 Flux (ϕ)

$$\phi = \phi_m \sin \omega t$$

Step:- 2 EMF

$$e = -N \frac{d\phi}{dt}$$

Step:-3 Find Max value of “e” per turn

For Per Turn, Take $N=1$

$$\begin{aligned} e &= - \frac{d\phi}{dt} \\ &= - \frac{d(\phi_m \sin \omega t)}{dt} \\ &= -\phi_m \omega \cos \omega t \end{aligned}$$

For Maximum Value, $\cos \omega t = \pm 1$

$$e_{max} = \phi_m \omega = 2\pi f \phi_m$$

Step:- 4 RMS vale

$$\begin{aligned} e_{rms} &= \frac{e_{max}}{\sqrt{2}} \\ &= \frac{2\pi f \phi_m}{\sqrt{2}} \end{aligned}$$

$$e_{rms} = 4.44 f \phi_m$$

Step :- 5 Primary and Secondary Induced EMF

➤ RMS Value of Primary Induced Voltage = $e_{rms} \times N_1$

$$E_1 = 4.44 f \phi_m N_1$$

➤ RMS Value of Secondary Induced Voltage = $e_{rms} \times N_2$

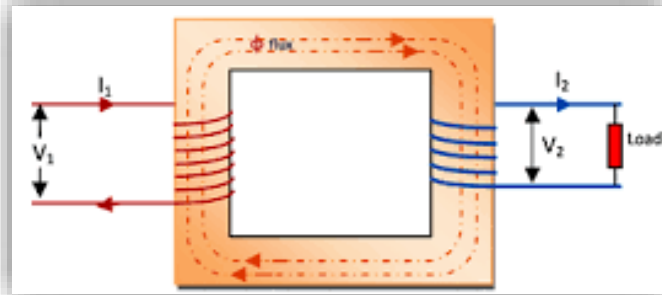
$$E_2 = 4.44 f \phi_m N_2$$

Transformer on load

$$E_1 = 4.44f\phi_m N_1$$

$$E_2 = 4.44f\phi_m N_2$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

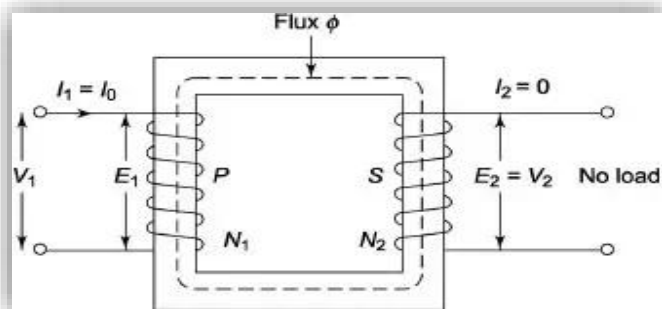


Transformer on No load

$$E_1 = V_1$$

$$E_2 = V_2$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{V_2}{V_1}$$



Transformation Ratio (K)

$$K = \frac{N_2}{N_1} = \frac{V_2}{V_1}$$

- If $K > 1 \Rightarrow N_2 > N_1 \Rightarrow V_2 > V_1 \Rightarrow$ **Step Up Transformer**
- If $K < 1 \Rightarrow N_2 < N_1 \Rightarrow V_2 < V_1 \Rightarrow$ **Step Down Transformer**
- If $K = 1 \Rightarrow N_2 = N_1 \Rightarrow V_2 = V_1 \Rightarrow$ **1:1 Transformer**

Turn Ratio (T)

$$T = \frac{N_1}{N_2} = \frac{\text{No of Primary winding Turns}}{\text{No of Secondary winding Turns}}$$

Voltage Ratio (T)

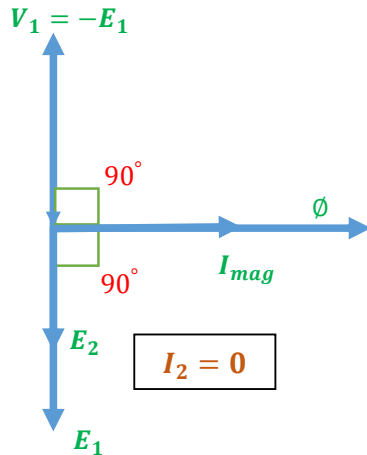
$$\frac{V_1}{V_2} = \frac{\text{Primary Voltage}}{\text{Secondary voltage}}$$

Current Ratio

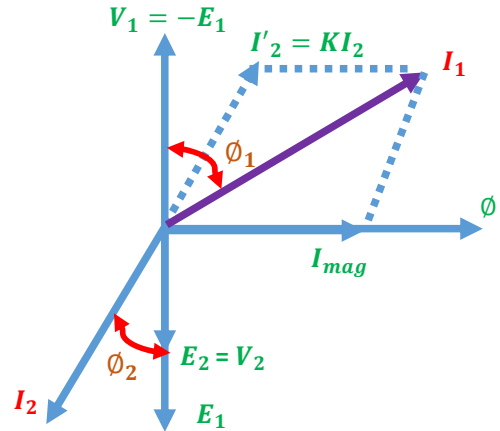
$$\frac{I_1}{I_2} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

Ideal Transformer

No Load

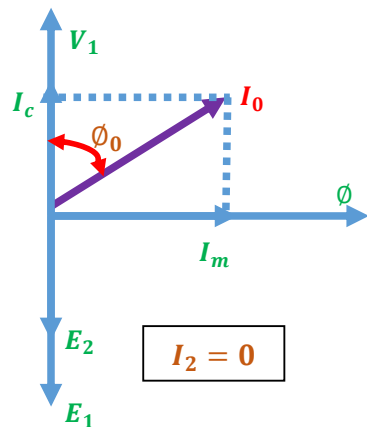


On Load



Practical Transformer

No Load



$$I_0 = \sqrt{I_m^2 + I_c^2}$$

$$I_m = I_0 \sin \phi_0$$
$$I_c = I_0 \cos \phi_0$$

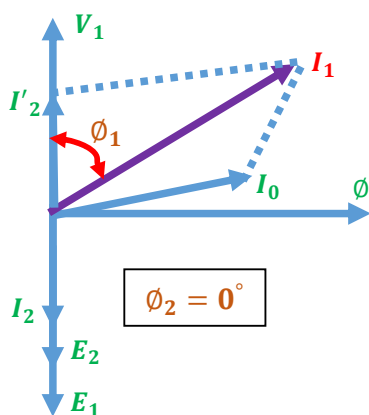
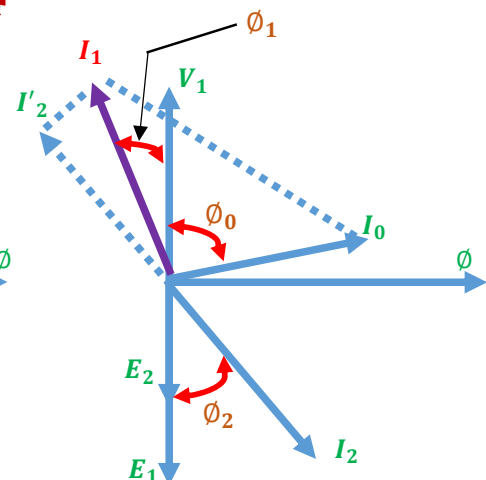
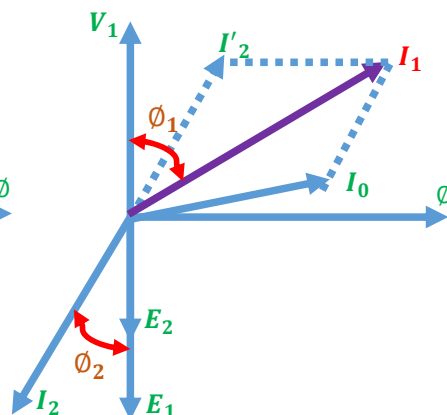
$$I_c = I_0 \cos \phi_0$$

- $I_0 = \text{No Load Current}$
- $I_m = \text{Magnetising Component}$
- $I_c = \text{Core Loss Component}$

$I_m = \text{Magnetising Component}$
 $I_c = \text{Core Loss Component}$

$$I_c = \text{Core Loss Component}$$

On Load


$$\phi_2 = 0^\circ$$


Resistive

Inductive

Capacitive

Auto Transformer

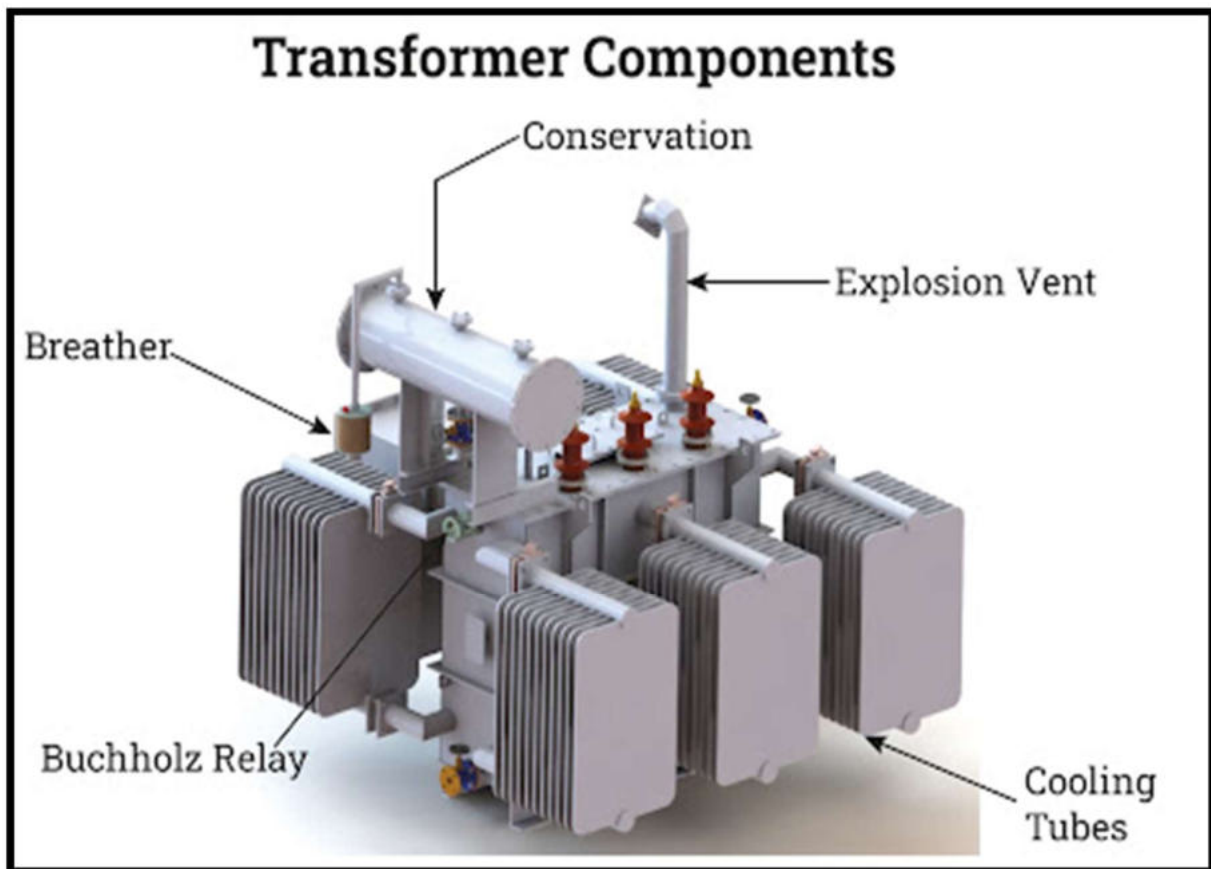


Single Phase



Three Phase

3 - Phase Transformer



PARTS OF A POWER TRANSFORMER



1. Transformer Tank
2. High Voltage Bushing
3. Low Voltage Bushing
4. Cooling Fins/Radiator
5. Cooling Fans
6. Conservator Tank
7. System Ground Terminal
8. Drain Valve
9. Breather
10. Oil Temperature/Pressure gauges
11. Bushing Current Transformers
12. Control Panel



Connection of 3-Phase Transformer

